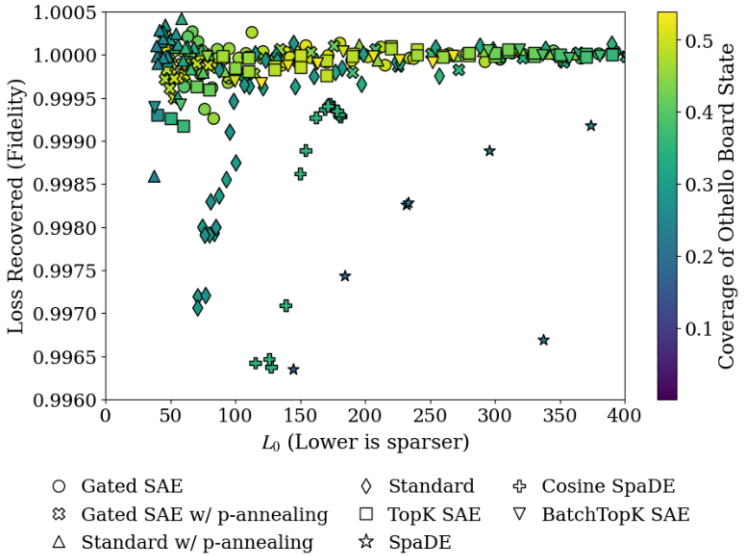
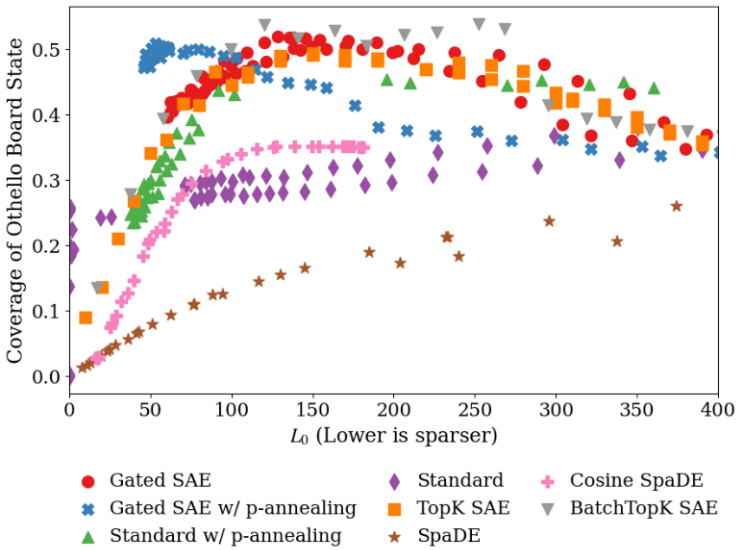
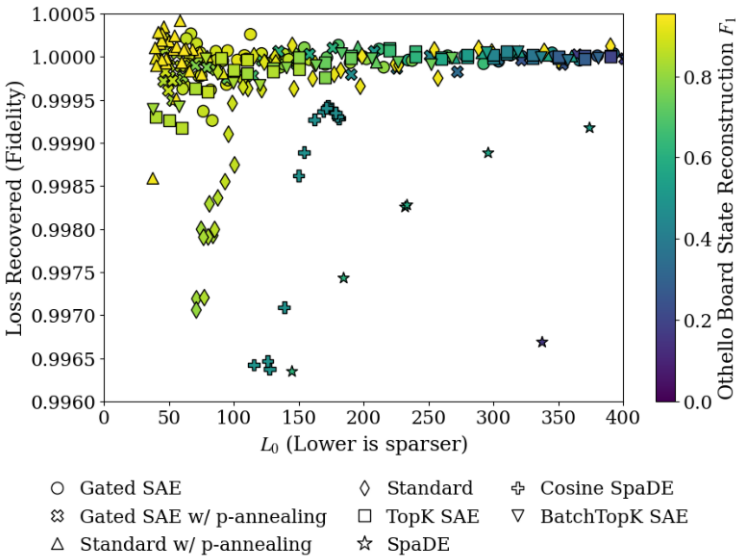

Loss Recovered (Fidelity)
1.0005
1.0000
0.9995
0.9990
0.9985
0.9980
0.9975
0.9970
0.9965
0.9960
0
50
100
150
200
250
300
350
400
L_0 (Lower is sparser)
Coverage of Othello Board State
0.5
0.4
0.3
0.2
0.1
Gated SAE
Gated SAE w/ p-annealing
Standard w/ p-annealing
Standard
TopK SAE
SpaDE
Cosine SpaDE
BatchTopK SAE

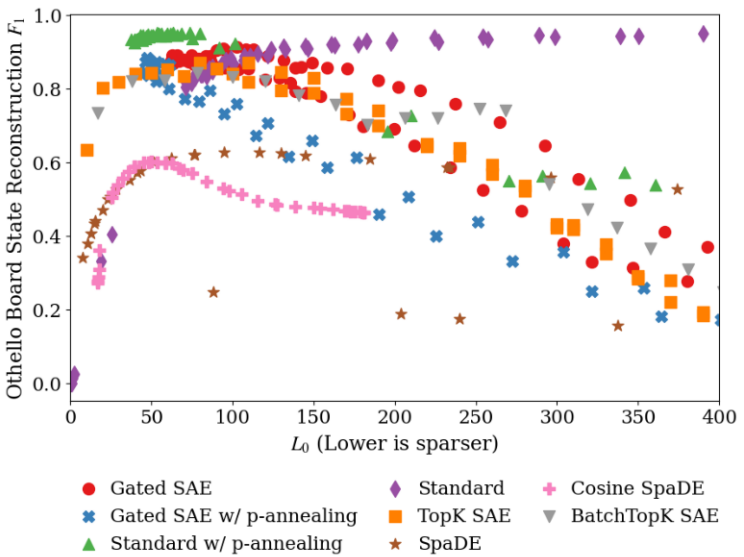
(a)


Coverage of Othello Board State
0.5
0.4
0.3
0.2
0.1
0.0
0
50
100
150
200
250
300
350
400
L_0 (Lower is sparser)
Gated SAE
Gated SAE w/ p-annealing
Standard w/ p-annealing
Standard
TopK SAE
SpaDE
Cosine SpaDE
BatchTopK SAE

(b)


Loss Recovered (Fidelity)
1.0005
1.0000
0.9995
0.9990
0.9985
0.9980
0.9975
0.9970
0.9965
0.9960
0
50
100
150
200
250
300
350
400
L_0 (Lower is sparser)
Othello Board State Reconstruction F_1
0.8
0.6
0.4
0.2
0.0
Gated SAE
Gated SAE w/ p-annealing
Standard w/ p-annealing
Standard
TopK SAE
SpaDE
Cosine SpaDE
BatchTopK SAE

(c)


Othello Board State Reconstruction F_1
1.0
0.8
0.6
0.4
0.2
0.0
0
50
100
150
200
250
300
350
400
L_0 (Lower is sparser)
Gated SAE
Gated SAE w/ p-annealing
Standard w/ p-annealing
Standard
TopK SAE
SpaDE
Cosine SpaDE
BatchTopK SAE

(d)